CS 5402: Introduction to Data Mining

SP2026

(Lecture -6)

Review & Bayes Classifier

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The Basics of Probability

Probability is a measure of the likelihood of an event, scaled from **0 to 1**.

- **0** = An impossible event (e.g., rolling a 7 on a standard six-sided die).
- **1** = A certain event (e.g., the sun will rise tomorrow).



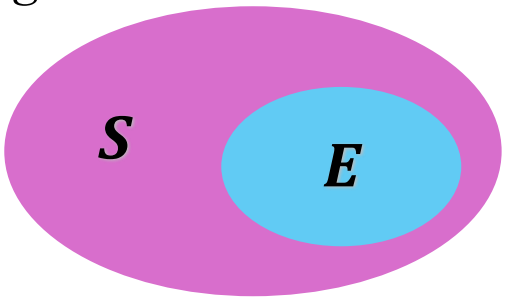


The Basics of Probability (Cont'd)

- **Experiment:** A procedure with a well-defined set of possible outcomes (e.g., flipping a coin, rolling a die).



- **Sample Space (S):** The set of **all possible outcomes**.
  - *Coin flip S:* {Heads, Tails}
  - *Die roll S:* {1, 2, 3, 4, 5, 6}
- **Event (E):** A specific outcome or a set of outcomes you care about.
  - *Die roll event "Rolling an even number":* E={2,4,6}



Basic Probability Rules

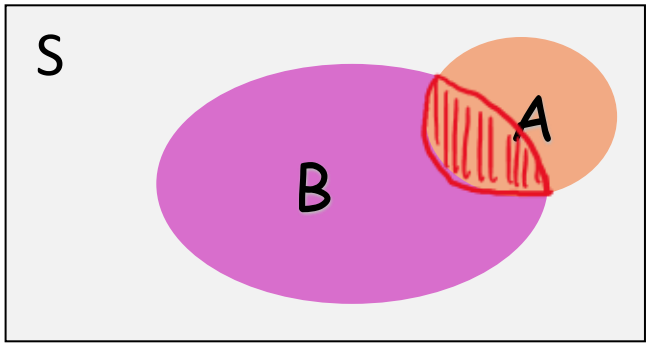
- **Calculating Simple Probability:**
  - $P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes in the sample space}}$
  - *Example:* $P(\text{Rolling an even number}) = \frac{|\{2,4,6\}|}{|\{1,2,3,4,5,6\}|} = \frac{3}{6} = 0.5$
- **Addition Rule ("OR"):** For mutually exclusive events (events that can't happen at the same time).
  - $P(A \text{ or } B) = P(A) + P(B)$
  - *Example:* $P(\text{Rolling a 1 or a 6}) = P(1) + P(6) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6}$
- **Multiplication Rule ("AND"):** For independent events (more on this next!).
  - $P(A \text{ and } B) = P(A) \times P(B)$
  - *Example:* $P(\text{Flipping Heads AND Rolling a 6}) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$

Conditional Probability P(A|B)

Conditional Probability is the likelihood of event A occurring, **given that** we know event B has already occurred.

Formula:

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

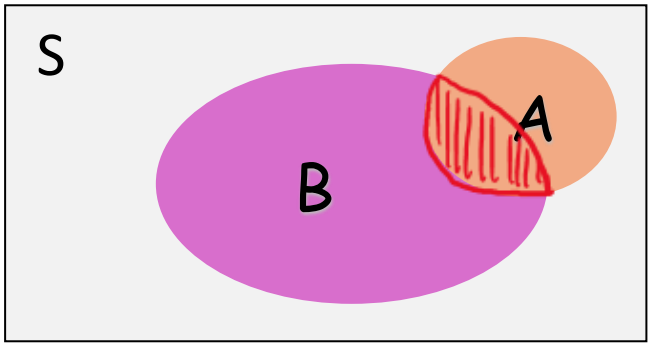


It answers the question:
"How does new information (B) change the probability of A?"

Example: Conditional Probability

Formula:

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$



Intuitive Example:

- Let's roll a single die.
  - The probability of rolling a 2 is P(2)=$\frac{1}{6}$.
- Now, what is the probability of rolling a 2 **given that we know the roll was an even number**?
  - Our sample space shrinks from {1,2,3,4,5,6} to just {2,4,6}.
  - Within this new sample space, the probability of rolling a 2 is now $\frac{1}{3}$.
  - P(Roll is 2 | Roll is Even)=$\frac{1}{3}$

Independent vs. Dependent Events

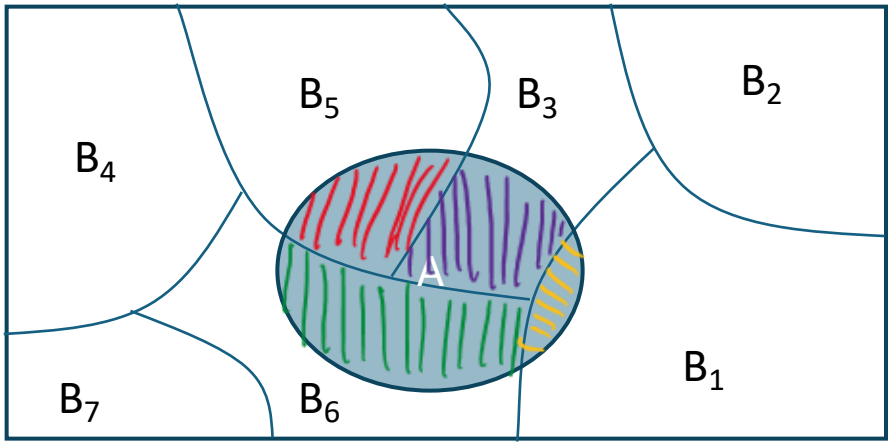
Independent Events: Knowing the outcome of one event gives you **no information** about the outcome of another.

- *Examples:* Flipping a coin twice, rolling two dice, drawing a card from a deck *with replacement*.
- Mathematically: P(A|B)=P(A).

Dependent Events: The outcome of one event **does** affect the outcome of another.

- *Example:* Drawing two cards from a deck *without replacement*. The first draw changes the probabilities for the second draw.
- P(GPA|Study_hours)

Rule of total probability



$$P(A) = \sum_{i=1}^{n} P(A|B_i) \cdot P(B_i)$$

$$P(A|B_1)\cdot P(B_1) + P(A|B_3)\cdot P(B_3) + P(A|B_5)\cdot P(B_5) + P(A|B_6)\cdot P(B_6) + 0 + \cdots + 0.$$

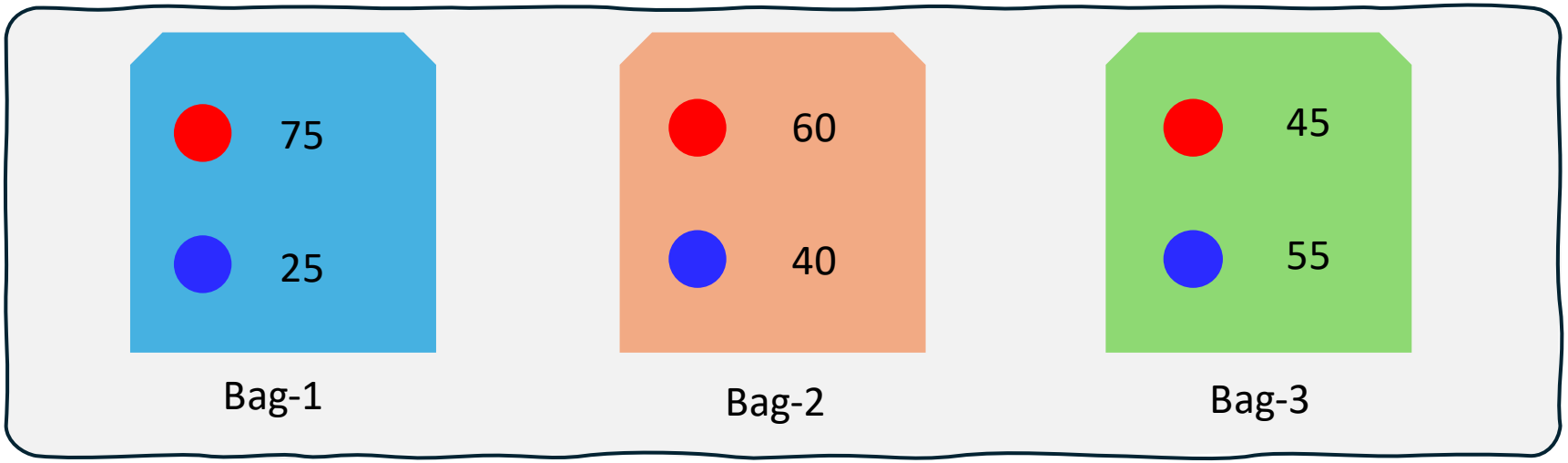
Naïve Bayes, GMM, Probabilistic prediction, Expectation, etc

Exercise-1

There are three bags that each contain 100 marbles:

- Bag 1 has 75 red and 25 blue marbles;
- Bag 2 has 60 red and 40 blue marbles;
- Bag 3 has 45 red and 55 blue marbles.

Randomly select one of the bags and then draw a marble from it at random. What is the probability of drawing a red marble?



Bayes' Theorem: The Heart of Inference

Bayes' Theorem is a powerful formula that lets us "reverse" conditional probabilities. If you know how data behaves given a certain state (P(B|A)), you can use it to find the probability of that state given the data (P(A|B)).

The Formula:

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$$

Bayes's Theorem

Provides a way to convert *a-priori* probabilities to *a-posteriori* probabilities:

$$P(A,B) = P(A|B)\,P(B)$$
$$P(B,A) = P(B|A)\,P(A)$$

The theorem:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A) * P(A)}{P(B)}$$

Likelihood (Evidence): How reliable is the test?

Prior (your starting belief): How common is the disease in the general population?

Posterior (Update your belief): Calculate the chance the patient truly has the disease based on the test result.

Marginalization (All possibilities): Consider all outcomes.

Bayes' Theorem: Example

The Classic Spam Filter: The goal is to determine the probability that an email is spam, given the words it contains.

- **Prior Belief (P(Spam)):** about 20% of all emails are spam. This is your **prior**.
- **Likelihood (P(”Great Deal" | Spam)):** You look at your historical data and find that this word appears in 5% of all spam emails. This is the **likelihood**. You also find it only appears in 0.01% of non-spam emails.
- **Posterior Belief (P(Spam | ”Great Deal")):** Now, you use Bayes' theorem to update your belief, a more informed probability—the **posterior**.
- **Evidence:** You receive a new email that contains the word ”Great Deal."

$$P(\text{ Spam } \mid \text{ “Great Deal” }) = \frac{P(\text{ “Great “Deal } \mid \text{ Spam }) \times P(\text{ Spam })}{P(\text{ “Great Deal” })}$$

Exercise-2:

How likely is it to have Disease-A(DA) if the test says so?

- Assume that roughly 0.1% of the population is infected.

$$p(X = \text{DA}) = 0.001$$

- The DA test reports positive for all infections.

$$p(Y = + \mid X = \text{DA}) = 1$$

- The DA test reports positive for 1% healthy people.

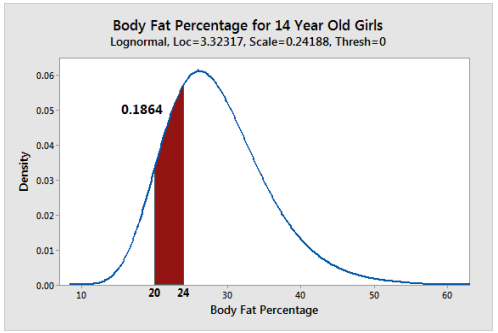
$$p(Y = + \mid X = \textit{healthy}) = 0.01$$

A patient tests positive. What is the probability they actually have the disease?

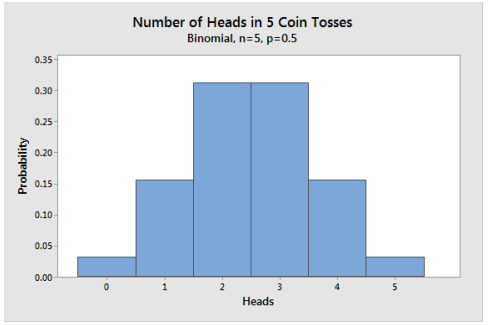
$$\mathrm{P}(\mathrm{X}=\mathrm{DA} \mid Y = +) \; ?$$

Random Variables

- A (scalar) random variable X is a function that maps the outcome of a random event into real scalar values
- There are two types of random variables
  - Continuous variable (take an infinite number of possible values.
    - Continuous random variables are usually measurements, examples include height, weight, the amount of sugar in an orange, the time required to run a mile, etc.
    - P($20 \leq x \leq 24$)=0.1864



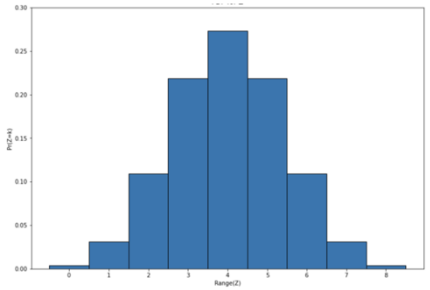
  - Discrete variable (take on only a countable number of distinct values, such as 0, 1, 2, 3,….)
    - Example: the total number of tails X you get if you flip coin 100 times; the number of children in a family; the number of defective light bulbs in a box of ten; etc.
    - P(X=2)=0.30



PDF, PMF and CDF

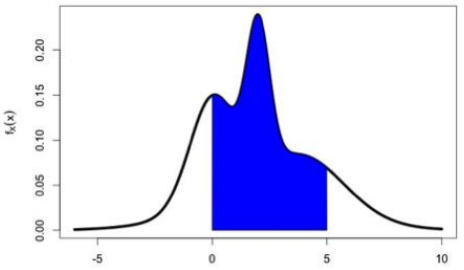
- Probability Mass Function (PMF): describe the probability distribution of the discrete random variable.

$$p(x) = P(X = x)$$



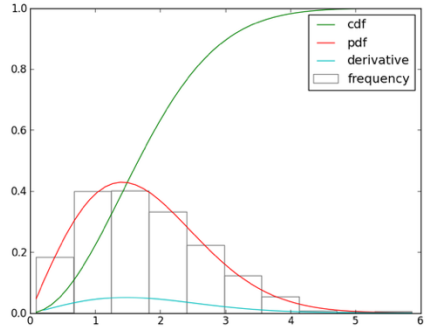
- Probability Density Function (PDF): describes the probability distribution of the continues random variable.

$$P(a \leq x \leq b) = \int_a^b f(x)dx$$



- Cumulative Probability Distribution (CDF): describe a random variable (continuous or discrete) will take on a value less than or equal to a certain value.

$$F_X(a) = P(x \leq \mathrm{a}) = \int_0^a f(x)dx$$

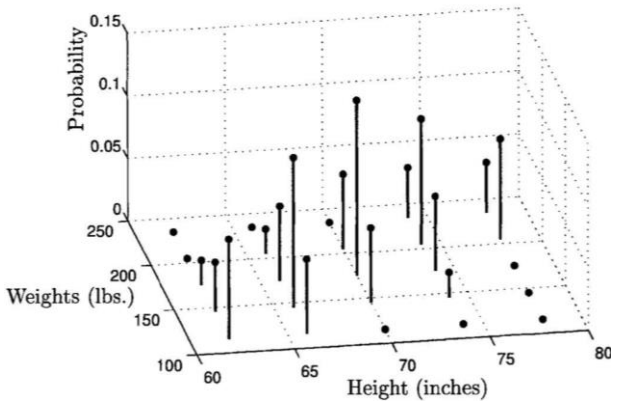


Joint Probability Distribution

- The Joint probability distributions of X and Y (both discrete) is defined by

$$p(x, y) = P(X = x, Y = y)$$

Satisfying (i) $p(x, y) \geq 0$; (ii) $\sum_{x,y} p(x, y) = 1$



- Also, the *marginal distributions* of X and Y can be derived from the joint distribution by summing (discrete) or integrating (continuous) over one variable.

$$p(x) = P(X = x) = \sum_{y} p(x, y)$$

$$p(y) = P(Y = y) = \sum_{x} p(x, y)$$

$$f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x, y)dy.$$

Exercise-3

Given the joint distribution of p(X, Y):

1) What is the marginal probability of P(X=1)?
2) What is the conditional probability of P(X=2|Y=1)?

X \ Y	1	2	3
1	0.32	0.03	0.01
2	0.06	0.24	0.02
3	0.02	0.03	0.27

Exercise-4

Given the joint distribution of p(X, Y):

1) What is the marginal probability of P(Y=3)?
2) What is the conditional probability of P(Y=2|X=2)?

X \ Y	1	2	3
1	0.32	0.03	0.01
2	0.06	0.24	0.02
3	0.02	0.03	0.27

Chain Rule

$$P(A,B) = P(A)P(B|A)$$

$$\begin{aligned} P(x,y,z) &= p(x)p(y|x)p(z|x,y) \\ &= p(z)p(y|z)p(x|y,z) \\ &= \cdots \end{aligned}$$

$$p(x_1,\ldots,x_N) = p(x_1)p(x_2 \mid x_1)\ldots p(x_N \mid x_1,\ldots,x_{N-1})$$

Probabilistic models (e.g., VAE), LLM, Reinforcement learning, Diffusion models, etc.

Mean and Variance

- Mean (Expectation):
  - Discrete RVs:

$$E(\mathrm{X}) = \sum_{x} xP(x)$$

$$E(g(X)) = \sum_{x} g(x)P(x)$$

  - Continuous RVs:

$$E(\mathrm{X}) = \int_{-\infty}^{+\infty} xf(x)dx$$

$$E(g(X)) = \int_{-\infty}^{+\infty} g(x)f(x)dx$$

Mean and Variance (Cont'd)

- Variance:
  - Discrete RVs:

$$\mathrm{Var}(\mathrm{X}) = E\left[(X-\mu)^2\right] = \sum_x (x-\mu)^2 P(x)$$

  - Continuous RVs:

$$\mathrm{Var}(\mathrm{X}) = E\left[(X-\mu)^2\right] = \int_{-\infty}^{+\infty} (x-\mu)^2 f(x)dx$$

- Covariance:

$$Cov(X,Y) = E((X-\mu_x)(Y-\mu_y)) = E(XY) - \mu_x\mu_y$$

Review of Information Theory

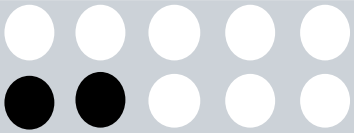
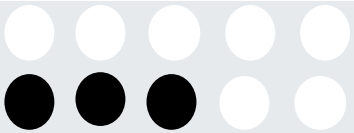
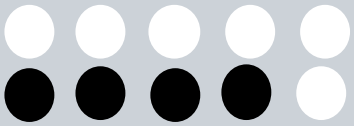
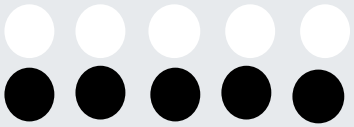
- Information Theory is a branch of applied mathematics and electrical engineering that deals with the quantification, storage, and communication of information.
- It was primarily developed by **Claude Shannon** in the mid-20th century and has since become fundamental in fields like data compression, cryptography, machine learning, and telecommunications.
- In simple terms, information theory is the study of how to measure, encode, and transmit information efficiently and securely
  - Information
  - Entropy
  - Mutual Information
  - etc

How do we measure information?

- Consider a discrete random variable X, the probability mass function (pmf): $p(x) = P(X = x)$
  - Denote $H(X)$ as the measure of uncertainty of X and is called Entropy.
  - Entropy H(X) is given as follows:

$$H(X) = -\sum_{x \in X} p(x) \log_2 p(x)$$

Entropy in Action:

(White, Black)	Confidence	Entropy
	★★★★★	0.000
	★★★★	0.469
	★★★	0.722
	★★	0.881
	★	0.971
		1.000

Entropy: Example-1

Example-1:

$$H(X) = -\sum_{x \in X} p(x) \log_2 p(x)$$

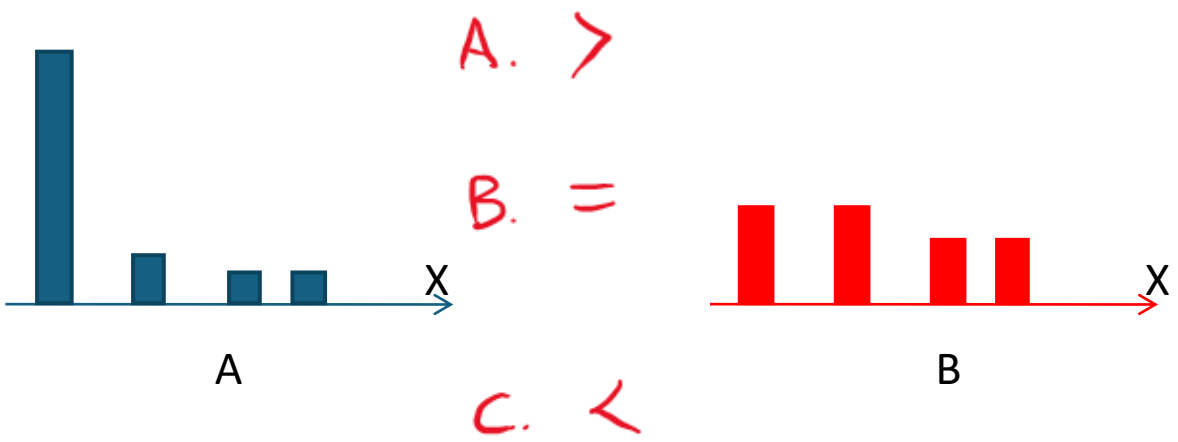
x	Sunny	cloudy	Rainy
$p(x)$	0.5	0.3	0.2

$$H(x) = -\left(0.5 \cdot \log_2 0.5 + 0.3 \cdot \log_2 0.3 + 0.2 \cdot \log_2 0.2\right) = 1.485$$

Entropy: Example-2

Example-2:

$$H(X) = -\sum_{x \in X} p(x) \log_2 p(x)$$



Joint Entropy

Let X and Y are discrete variables and $(X, Y) \sim p(x, y)$, then the joint entropy of X and Y is:

$$H(X, Y) = -\sum_{x \in X} \sum_{y \in Y} p(x, y) \log_2 p(x, y)$$

The joint entropy represents the amount of information needed on average to specify the value of two discrete random variables.

Conditional Entropy

Let X and Y are discrete variables. p(x, y) and p(x|y) are the values of their joint and conditional probability distributions, then the conditional entropy of X given Y is:

$$H(X|Y) = -\sum_{x \in X} \sum_{y \in Y} p(x, y) \log_2 p(x|y)$$

- The conditional entropy measures the amount of uncertainty (entropy) left in a random variable X when we already know the value of another random variable Y.
- It tells us how much additional information you still need to supply on average to describe X once Y is given.

Conditional Entropy (Cont'd)

Let X and Y are discrete variables. p(x, y) and p(x|y) are the values of their joint and conditional probability distributions, then the conditional entropy of X given Y is:

$$H(X|Y) = -\sum_{x \in X} \sum_{y \in Y} p(x,y) \log_2 p(x|y)$$

$$-\sum_x \sum_y p(x,y) \log_2 \frac{p(x,y)}{p(y)} = -\sum_x \sum_y p(x,y) \cdot \log_2 p(x,y) \rightarrow H(X,Y)$$

$$+ \sum_x \sum_y p(x,y) \cdot \log_2 p(y)$$

$$\sum_y p(y) \cdot \log_2 p(y) \rightarrow -H(Y)$$

$$H(X|Y) = H(X,Y) - H(Y)$$

Mutual Information: MI

- Remember independence?
  - If X⊥Y then knowing Y won't change our belief about X
  - But what if X and Y are not independent, how can we measure the level of dependence?
- Mutual information of two random variables X and Y, whose joint distribution is defined by $p(x, y)$ is given by

$$I(X;Y) = \sum_{y}\sum_{x} p(x,y)\log\left(\frac{p(x,y)}{p(x)p(y)}\right)$$

$$I(X;Y) = H(X) - H(X|Y)$$

OR

$$I(X;Y) = H(Y) - H(Y|X)$$

  - "The amount of uncertainty in X which is removed by knowing Y"
  - Symmetric: $I(X;Y) = I(Y;X)$
  - I(X;Y) = 0 iff, X and Y are independent!

Mutual Information: Examples

- Exp-1: Coin Toss
  - Toss two fair coins:
    - X is the result of the firs coin (Heads or Tails)
    - Y is the result of the second coin (Heads or Tails)
  - Since the two coins tosses are independent, knowing X does not help you predict Y.

$$I(X; Y) = 0$$

- Exp-2: Student Exam Scores and Studying Time
  - Let X be the hours a student studies.
  - Let Y be the exam score.

  If students who study more tend to score higher, knowing how long a student studied (X) helps predict their score (Y).

$$I(X; Y) > 0$$

Relative Entropy: Motivation

- Imagine you live in a city where the probability of a rainy day (P) is:

Rainy	Sunny
0.7	0.3

- Now, your friend, who has never visited your city, makes a prediction (Q):

Rainy	Sunny
0.5	0.5

Question: how bad or good your friend's prediction (Q) is compared to the actual distribution (P)?

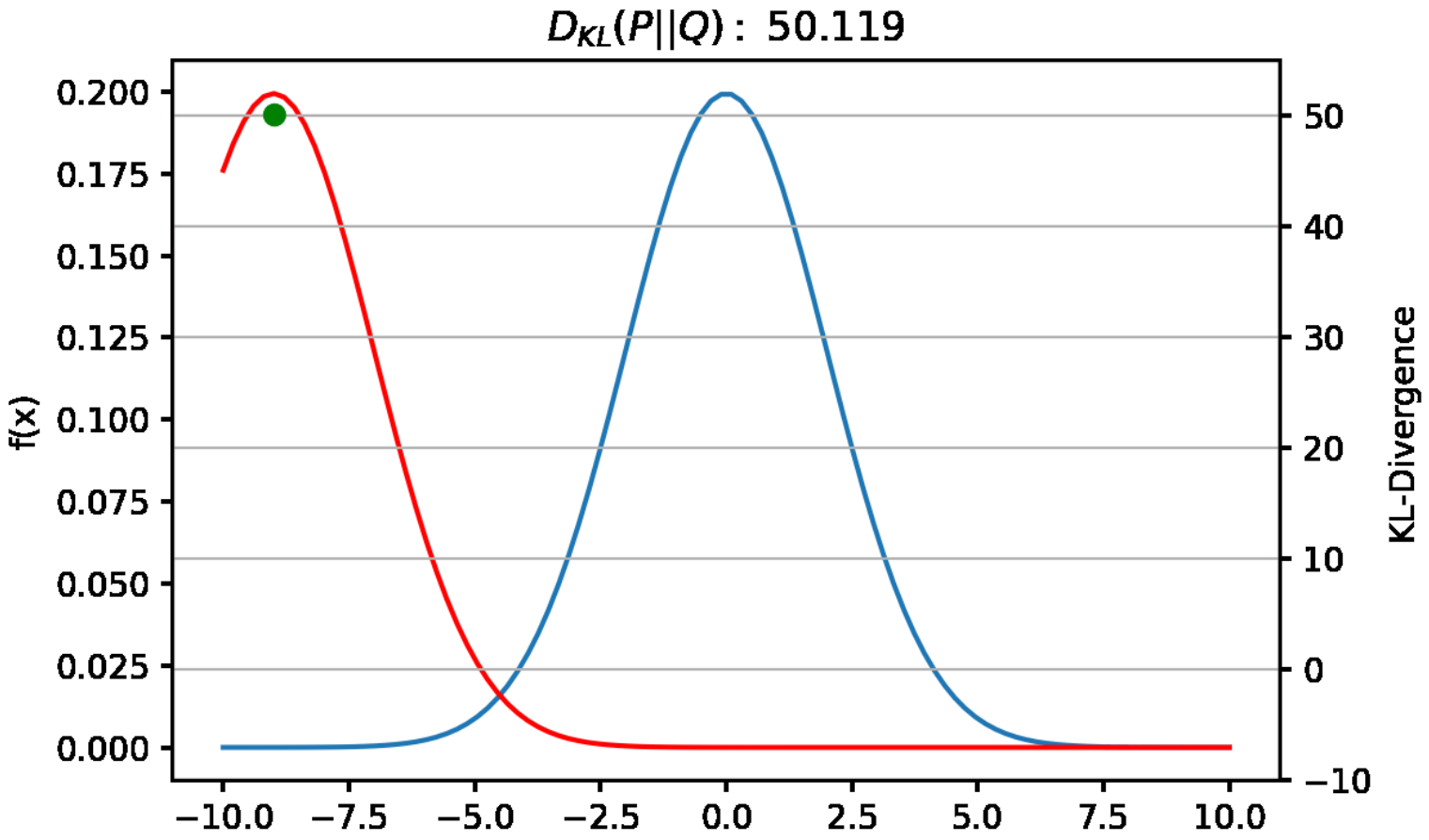
Relative Entropy

- Relative Entropy, also known as Divergence or Kullback-Leibler (KL) distance, is a statistical measurement from information theory that is commonly used to quantify the difference between one probability distribution from a reference probability distribution.
- Let $p(x)$ and $q(x)$ are two probability distribution of a discrete random variable x, the KL divergence is defined as:

$$D_{KL}(p(x)||q(x)) = \sum_{x \in X} p(x) \log_2 \frac{p(x)}{q(x)}$$

  - Measure the dissimilarity of two probability distributions.
  - Relative entropy is not symmetric. In other words, it is not necessarily true that $D_{KL}(p(x)||q(x)) = D_{KL}(q(x)||p(x))$
  - Relative entropy is always non-negative. $D_{KL}(p(x)||q(x)) \geq 0$

KL divergence example



Bayes Classifier

Let the training dataset $D = \{(x_1, y_1), (x_{2,}, y_2), \ldots (x_N, y_N)\}$, $x_i \in \mathcal{R}^d$, $y_i \in \{c_1, c_2, \ldots, c_k\}$.

The Bayes classifier estimates the posterior probability $P(c_i|x)$ for each class c_i, and chooses the class that has the largest probability. The predicted class for x is given as:

$$\hat{y} = arg \max_{c_i}\{P(c_i|x)\}$$

According to the Bayes theorem, we have:

$$P(c_i|x) = \frac{P(x|c_i)P(c_i)}{P(x)}$$

Because $P(x)$ is fixed for a given point, Bayes rule can be rewritten as:

$$\hat{y} = arg \max_{c_i}\{P(c_i|x)\} = arg \max_{c_i}\left\{\frac{P(x|c_i)P(c_i)}{P(x)}\right\} = arg \max_{c_i}\{P(x|c_i)P(c_i)\}$$

Estimating the Prior Probability: $P(c_i)$

Let D_i denote the subset of points in D that are labeled with class c_i:

$$D_i = \{x_j \in D \,|\, x_j has\ class\ y_j = c_j\}$$

Let the size of the dataset D be given as $|D| = N$, and let the size of each class-specific subset D_i be given as $|D_i| = n_i$.

The prior probability for class c_i can be estimated as follows:

$$\hat{P}(c_i) = \frac{|D_i|}{|D|} = \frac{n_i}{N}$$

Estimating the Likelihood: $P(x|c_i)$

Given training dataset $D = \{(x_1, y_1), (x_{2,}, y_2), \dots (x_N, y_N)\}$, $x_i \in \mathcal{R}^d$, $y_i \in \{c_1, c_2, \dots, c_k\}$.

$$x_i = \left(x_i^1, x_i^2, \dots, x_i^d\right)^T \in \mathcal{R}^d$$

How many parameters for $P(x_i|c_i) = P(x_i^1, x_i^2, \dots, x_i^d|c_i)$

Assume that $x_i^j \in \{0, 1\}$

2^d

Naïve Bayes

The Naïve Bayes approach makes the simple assumption that *all the attributes are conditionally independent*, which implies that the likelihood can be decomposed into a product of dimension-wise probabilities:

$$P(x_i|c_i) = P\left(x_i^1, x_i^2, \ldots, x_i^d | c_i\right) = \prod_{j=1}^{d} P(x_i^j|c_i)$$

2^d $2d$

Reduce the number of parameters to estimate $P(x|c_i)$

This assumption is usually false, but surprisingly effective!

Naïve Bayes Algorithm – discrete attribute x_i^j

Let categorical attribute $x_i^j \in \{a_1, a_2, \ldots, a_l\}$, for each value a_k:

$$P(x_i^j = a_k | c_i) = \frac{|D_{x_i^j = a_k} \wedge D_i|}{|D_i|} = \frac{n_{ik}}{n_i}$$

Number of items in dataset D for which $x_i^j = a_k$ and $y = c_i$

Naïve Bayes Algorithm – discrete attribute x_i^j (Cont'd)

Let categorical attribute $x_i^j \in \{a_1, a_2, \ldots, a_l\}$, for each value a_k:

$$P(x_i^j = a_k | c_i) = \frac{|D_{x_i^j = a_k} \wedge D_i|}{|D_i|}$$

Classify x_{new}:

$$y_{new} = arg \max_{c_i} \left\{ P(c_i) \prod_{j=1}^{d} P(x_{new}^j | c_i) \right\}$$

In practice (for numerical stability):

$$y_{new} = \arg\max_{c_i} \left[\log P(c_i) + \sum_{j=1}^{d} \log P\left(x_{new}^j \mid c_i\right) \right]$$

From Discrete Attribute to Continuous Attribute

- Convert to discrete attributes:
  - Discretization: partition the range into bins:
    - Replace continuous value with bin value. (attribute changed from continuous to ordinal)

- Probability density estimation:
  - Assume the attribute follows a distribution f_θ (i.e., normal distribution)
  - Use data to estimate parameters θ of distribution (e.g., mean and standard deviation)
  - Once probability distribution is known, use it to estimate the conditional probability $P(x_i^j | c_i)$

Bayes Classifier with Continuous Attribute: Example

X_i=	$(x_i^1, \ldots$	x_i^j	$\ldots, x_i^d)$	Y
x_1	…	168.2	…	+
x_2	…	175.8	…	-
x_3	…	180.3	…	-
x_4	…	166.5	…	+
x_5	…	174.1	…	+
x_6	…	179.3	…	+
x_7	…	183.2	…	-
x_8	…	176.8	…	-
…	…	…	…	…
x_{10}	…	181.6	…	-

- Normal distribution:

$$P(x_i^j | c_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x_i^j - \mu)^2}{2\sigma^2}}$$

- Learn parameters:
  - If class="+": $P(x_i^j | " + ")$
  - If class="-": $P(x_i^j | " - ")$

Bayes Classifier: Example (1)

Samples

Features	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	X_{11}	X_{12}	X_{13}	X_{14}	X_{15}
$X^{(1)}$	1	1	1	1	1	2	2	2	2	2	3	3	3	3	3
$X^{(2)}$	S	M	M	S	S	S	M	M	L	L	L	M	M	L	L
Y	-1	-1	1	1	-1	-1	-1	1	1	1	1	1	1	1	-1

$$X_i = \left(X_i^{(1)}, X_i^{(2)}\right), \quad X_i^{(1)} \in \{1,2,3\}, \quad X_i^{(2)} \in \{S,M,L\}, \quad Y \in \{1,-1\}$$

What is the class label Y for new $\mathrm{x} = (2,S)^T$?

$$P(Y=1|X=x) \quad \begin{matrix} \geq \\ = \\ < \end{matrix} \quad P(Y=-1|X=x)$$

Bayes Classifier: Example (2)

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	X_{11}	X_{12}	X_{13}	X_{14}	X_{15}
$X^{(1)}$	1	1	1	1	1	2	2	2	2	2	3	3	3	3	3
$X^{(2)}$	S	M	M	S	S	S	M	M	L	L	L	M	M	L	L
Y	-1	-1	1	1	-1	-1	-1	1	1	1	1	1	1	1	-1

$$X_i = \left(X_i^{(1)}, X_i^{(2)}\right), \qquad X_i^{(1)} \in \{1,2,3\}, \qquad X_i^{(2)} \in \{S,M,L\}, \qquad Y \in \{1,-1\}$$

What is the class label Y for new $\mathrm{x} = (2,S)^T$?

$$P(Y=1|X=x) \quad \geq \\ = \\ < \quad P(Y=-1|X=x)$$

$$= \frac{P(Y=1)P(X=x|Y=1)}{P(X=x)} \qquad = \frac{P(Y=-1)P(X=x|Y=-1)}{P(X=x)}$$

Bayes Classifier: Example (3)

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	X_{11}	X_{12}	X_{13}	X_{14}	X_{15}
$X^{(1)}$	1	1	1	1	1	2	2	2	2	2	3	3	3	3	3
$X^{(2)}$	S	M	M	S	S	S	M	M	L	L	L	M	M	L	L
Y	-1	-1	1	1	-1	-1	-1	1	1	1	1	1	1	1	-1

$$X_i = \left(X_i^{(1)}, X_i^{(2)}\right), \qquad X_i^{(1)} \in \{1,2,3\}, \qquad X_i^{(2)} \in \{S,M,L\}, \qquad Y \in \{1,-1\}$$

$$P(Y=1) = \frac{9}{15}, \qquad P(Y=-1) = \frac{6}{15}$$

$$P\left(X^{(1)}=1 \middle| Y=1\right) = \frac{2}{9}, \qquad P\left(X^{(1)}=2 \middle| Y=1\right) = \frac{3}{9}, \qquad P\left(X^{(1)}=3 \middle| Y=1\right) = \frac{4}{9}$$

$$P\left(X^{(2)}=S \middle| Y=1\right) = \frac{1}{9}, \qquad P\left(X^{(2)}=M \middle| Y=1\right) = \frac{4}{9}, \qquad P\left(X^{(2)}=L \middle| Y=1\right) = \frac{4}{9}$$

$$P\left(X^{(1)}=1 \middle| Y=-1\right) = \frac{3}{6}, \qquad P\left(X^{(1)}=2 \middle| Y=-1\right) = \frac{2}{6}, \qquad P\left(X^{(1)}=3 \middle| Y=-1\right) = \frac{1}{6}$$

$$P\left(X^{(2)}=S \middle| Y=-1\right) = \frac{3}{6}, \qquad P\left(X^{(2)}=M \middle| Y=-1\right) = \frac{2}{6}, \qquad P\left(X^{(2)}=L \middle| Y=-1\right) = \frac{1}{6}$$

Bayes Classifier: Example (4)

No optimization, no gradient descent!

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	X_{11}	X_{12}	X_{13}	X_{14}	X_{15}
$X^{(1)}$	1	1	1	1	1	2	2	2	2	2	3	3	3	3	3
$X^{(2)}$	S	M	M	S	S	S	M	M	L	L	L	M	M	L	L
Y	-1	-1	1	1	-1	-1	-1	1	1	1	1	1	1	1	-1

$$X_i = \left(X_i^{(1)}, X_i^{(2)}\right), \quad X_i^{(1)} \in \{1,2,3\}, \quad X_i^{(2)} \in \{S,M,L\}, \quad Y \in \{1,-1\}$$

What is the class label Y for new $x = (2,S)^T$?

Therefore, $Y = -1 \; for \; x = (2,S)^T$

Assume that $X^{(1)} \perp X^{(2)}$

- $P(Y=1)P(X=x|Y=1) = P(Y=1)P\left(X^{(1)}=2\middle|Y=1\right)P\left(X^{(2)}=S\middle|Y=1\right)$
$$= \frac{9}{15} * \frac{3}{9} * \frac{1}{9} = \frac{1}{45}$$
- $P(Y=-1)P(X=x|Y=-1) = P(Y=-1)P\left(X^{(1)}=2\middle|Y=-1\right)P\left(X^{(2)}=S\middle|Y=-1\right)$
$$= \frac{6}{15} * \frac{2}{6} * \frac{3}{6} = \frac{1}{15}$$

Summary

Pros:

- Needs **very little data**
- Robust to **irrelevant features**
- Works well in **high-dimensional sparse spaces** (text)
- Fast training & prediction
- Often a **strong baseline**

- Cons:
  - Independence assumption often violated
  - Cannot model feature interactions
  - Poor probability calibration
  - Struggles with correlated features

Any Question?